Object Detection with YOLO11: what changed

Simone Bonato · 9 November 2026

Your YOLOv7 workbook, modernized as you asked: the same storyline (setup, data, training, predictions, TP/FP/FN, optional experiments) and your tasks with Sample Answers, on YOLO11 and the Ultralytics Python API, with a new dataset and two new interactive parts. Each page shows your version on the left and the new one on the right; the red numbers mark the exact place.

Is it shorter? In cells and commands, yes: 66 cells instead of 90, and 1 shell command instead of 23. In reading, no: about 2,600 words of text instead of 1,500, because the precision–recall, AP and NMS parts are new and explained. If it should be shorter still, the first candidates to cut are the AP-per-class figure (page 7) and the freezing experiment (page 9).

Tested: runs top to bottom headless (CPU smoke test, every check passing) and once in full on a Mac: YOLO11n trained 20 epochs in 7.5 minutes there, mAP@50 0.83, mAP@50–95 0.50; the whole notebook including both optional experiments ran in 45 minutes without an error. A full run in Colab on a T4 is still to do. Why YOLO11 and not the newer YOLO26: on Ultralytics' own 37-dataset fine-tuning benchmark YOLO11n scores slightly higher (46.4 vs 44.4 mAP@50–95), and its classic pipeline shows the NMS step the slider explores.

1 · Setup: from version pins to one install

Your workbook had to pin an old Colab runtime (Python 3.11, 2025.07), reinstall torch 2.5.0 and clone the YOLOv7 repository. Now one cell installs Ultralytics (pinned 8.4.208) on Colab's current runtime, and training, validation and prediction are each one Python call. Drive and Weights & Biases are no longer needed; the training curves are drawn in the notebook.

Your YOLOv7 workbook

1 The current configuration runs only on Python 3.11. Therefore, we need to select an older runtime in Colab. Runtime → Change runtime type → Runtime version → 2025.07 → Save

```
import torch
print(torch.__version__)
```

```
2.8.0+cu126
```

```
!pip uninstall -y torch torchvision torchaudio
```

```
Found existing installation: torch 2.8.0+cu126
Uninstalling torch-2.8.0+cu126:
  Successfully uninstalled torch-2.8.0+cu126
Found existing installation: torchvision 0.23.0+cu126
Uninstalling torchvision-0.23.0+cu126:
  Successfully uninstalled torchvision-0.23.0+cu126
Found existing installation: torchaudio 2.8.0+cu126
Uninstalling torchaudio-2.8.0+cu126:
  Successfully uninstalled torchaudio-2.8.0+cu126
```

```
!pip install torch==2.5.0 torchaudio==2.5.0 torchvision==0.20.0
```

2

```
Collecting torch==2.5.0
  Downloading torch-2.5.0-cp312-cp312-manylinux1_x86_64.whl.metadata (28 kB)
Collecting torchaudio==2.5.0
  Downloading torchaudio-2.5.0-cp312-cp312-manylinux1_x86_64.whl.metadata (6.4 kB)
Collecting torchvision==0.20.0
  Downloading torchvision-0.20.0-cp312-cp312-manylinux1_x86_64.whl.metadata (6.1 kB)
Requirement already satisfied: filelock in /usr/local/lib/python3.12/dist-packages (from torch==2.5.0) (3.20.0)
Requirement already satisfied: typing-extensions>=4.8.0 in /usr/local/lib/python3.12/dist-packages (from torch==2.5.0) (4.15.0)
Requirement already satisfied: networkx in /usr/local/lib/python3.12/dist-packages (from torch==2.5.0) (3.5)
Requirement already satisfied: jinja2 in /usr/local/lib/python3.12/dist-packages (from torch==2.5.0) (3.1.6)
Requirement already satisfied: fsspec in /usr/local/lib/python3.12/dist-packages (from torch==2.5.0) (2025.3.0)
Collecting nvidia-cuda-nvrtc-cu12==12.4.127 (from torch==2.5.0)
  Downloading nvidia_cuda_nvrtc_cu12-12.4.127-py3-none-manylinux2014_x86_64.whl.metadata (1.5 kB)
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  Downloading nvidia_cuda_cupti_cu12-12.4.127-py3-none-manylinux2014_x86_64.whl.metadata (1.6 kB)
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  Downloading nvidia_cudnn_cu12-9.1.0.70-py3-none-manylinux2014_x86_64.whl.metadata (1.6 kB)
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  Downloading nvidia_cublas_cu12-12.4.5.8-py3-none-manylinux2014_x86_64.whl.metadata (1.5 kB)
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  Downloading nvidia_cufft_cu12-11.2.1.3-py3-none-manylinux2014_x86_64.whl.metadata (1.5 kB)
Collecting nvidia-curand-cu12==10.3.5.147 (from torch==2.5.0)
  Downloading nvidia_curand_cu12-10.3.5.147-py3-none-manylinux2014_x86_64.whl.metadata (1.5 kB)
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  Downloading nvidia_cusolver_cu12-11.6.1.9-py3-none-manylinux2014_x86_64.whl.metadata (1.6 kB)
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  Downloading nvidia_cusparse_cu12-12.3.1.170-py3-none-manylinux2014_x86_64.whl.metadata (1.6 kB)
Collecting nvidia-nccl-cu12==2.21.5 (from torch==2.5.0)
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Requirement already satisfied: setuptools in /usr/local/lib/python3.12/dist-packages (from torch==2.5.0) (75.2.0)
Collecting sympy==1.13.1 (from torch==2.5.0)
  Downloading sympy-1.13.1-py3-none-any.whl.metadata (12 kB)
Requirement already satisfied: numpy in /usr/local/lib/python3.12/dist-packages (from torchvision==0.20.0) (2.0.2)
Requirement already satisfied: pillow!=8.3.*,>=5.3.0 in /usr/local/lib/python3.12/dist-packages (from torchvision==0.20.0) (11.3.0)
Requirement already satisfied: mpmath<1.4,>=1.1.0 in /usr/local/lib/python3.12/dist-packages (from sympy==1.13.1->torch==2.5.0) (1.3.
0)
Requirement already satisfied: MarkupSafe>=2.0 in /usr/local/lib/python3.12/dist-packages (from jinja2->torch==2.5.0) (3.0.3)
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Downloading nvidia_nvjitlink_cu12-12.4.127-py3-none-manylinux2014_x86_64.whl (21.1 MB)
```

New

Setup

1 The next cell installs Ultralytics (the library that ships YOLO11), checks for a GPU, and loads the helpers the notebook uses. It takes about a minute the first time.

```
# ▸ Setup (run me first)
```

```
No GPU found. Training will be very slow: Runtime → Change runtime type → T4 GPU, then run again.
Colab: False · smoke mode: False · device: cpu
```

New

YOLO11 comes in five sizes. Larger models detect more accurately but need more memory and time. The numbers below are Ultralytics' published results on COCO at 640 px (mAP@50–95 is explained in the section *Precision, Recall and AP*).

3

Model	Parameters	Compute (GFLOPs)	COCO mAP@50–95	Typical target
YOLO11n	2.6 M	6.5	39.5	phones, edge devices
YOLO11s	9.4 M	21.5	47.0	small GPUs
YOLO11m	20.1 M	68.0	51.5	consumer GPUs
YOLO11l	25.3 M	86.9	53.4	cloud GPUs
YOLO11x	56.9 M	194.9	54.7	high-end cloud GPUs

We fine-tune **YOLO11n**, the smallest, so that training takes minutes. The optional experiments at the end compare it with YOLO11s.

Your YOLOv7 workbook

Download YOLO Repository

3

```
# only needs to be executed once or in case an update is desired/required
# this is the official YOLOv7
!git clone https://github.com/WongKinYiu/yolov7.git
```

```
Cloning into 'yolov7'...
remote: Enumerating objects: 1197, done.
remote: Total 1197 (delta 0), reused 0 (delta 0), pack-reused 1197 (from 1)
Receiving objects: 100% (1197/1197), 74.29 MiB | 18.49 MiB/s, done.
Resolving deltas: 100% (511/511), done.
```

2 · A new dataset, from outside COCO

Cattle from a drone became weeds and crop plants from a drone (Roboflow-100 "weed-crop-aerial", CC BY 4.0, downloaded from Hugging Face without a login). We train on a fixed subset of 500 training and 150 validation images. A new first task shows what the COCO model predicts before fine-tuning, which motivates the training.

Your YOLOv7 workbook

1 For this exercise, we are working with the cattle dataset (data, official website). The cattle dataset was created to detect and count cattle using *unmanned aerial vehicle* (UAV) images. For this exercise, cattle dataset1 (653 images) was used as training data, and dataset2 (14 images) as validation data. To keep the training times shorter during the deep dive, we downsized the images to 800x600px.

Further image preprocessing steps such as scaling of the pixel values could be applied, as illustrated in the segmentation workbook. For simplicity, we are not performing these preparation steps here.

New

The Dataset

Download the dataset, recall YOLO's label format, and take a short look at the images and their boxes.

1 A drone flies low over a field and takes photos. A farmer who knows where the weeds are can spray only those spots instead of the whole field, which saves herbicide and money. For this, a detector has to find every weed, and tell it apart from the crop plants around it.

Our dataset has photos taken from above, looking down at the soil of crop fields, with boxes around crop plants and weeds: two classes, and almost all boxes are weeds. The objects are tiny, around 20 pixels wide in a 640 × 640 image, and look nothing like COCO's everyday photos.

You explored a detection dataset in depth in *Know Your Images*. Here we only look briefly, enough to know what the model has to learn.

New

```
# ▸ 🔎 Browse Training Images with Their Boxes
```

None

New

2 🟢 Before Fine-Tuning: What Does a COCO Model See?

The pretrained YOLO11n has learned COCO's 80 everyday classes: people, cars, dogs, cups, and so on. None of them is one of our classes.

Before running the code:

- What will the pretrained model predict on our images: nothing, wrong classes, or our objects under another name?

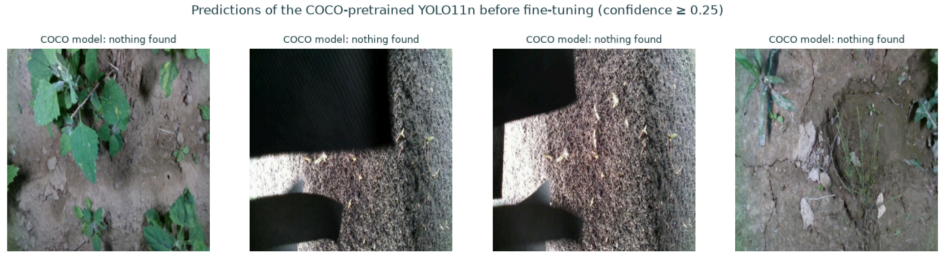
Explore:

- Run the cell. It shows the predictions of the COCO model on four validation images.

Observe and interpret:

- Which COCO classes does the model use, if any? What does this tell you about why we need to fine-tune?

```
# ▸ 🔎 Predict with the COCO-Pretrained Model
```



3 · Dataset exploration: one short look

Your qualitative and quantitative analysis tasks are covered by Know Your Images, so here they are replaced by one sentence and one image browser.

Your YOLOv7 workbook

Analyze the Data Qualitatively

Exercise

1. Analyze the images and labels by visual inspection and report the problems.
2. Get a qualitative sense of the data and try to find any sort of patterns that are visible to the human brain

Your YOLOv7 workbook

Analyze the Data Quantitatively

Exercise

2. Analyze the images and labels using code. E.g, count number of images, images shapes and their frequencies, classes per image, numbers and sizes of object boxes, average +/- std of number of pixels, pixel value ranges and distributions, imbalances, bias, and other quantitative measures that you can think of.

New

The Dataset

Download the dataset, recall YOLO's label format, and take a short look at the images and their boxes.

1. A drone flies low over a field and takes photos. A farmer who knows where the weeds are can spray only those spots instead of the whole field, which saves herbicide and money. For this, a detector has to find every weed, and tell it apart from the crop plants around it.

Our dataset has photos taken from above, looking down at the soil of crop fields, with boxes around crop plants and weeds: two classes, and almost all boxes are weeds. The objects are tiny, around 20 pixels wide in a 640 × 640 image, and look nothing like COCO's everyday photos.

1. You explored a detection dataset in depth in *Know Your Images*. Here we only look briefly, enough to know what the model has to learn.

4 · Training: your batch-size and epochs task kept

Same task, now as two sliders with a Sample Answer, followed by the training curves (loss and validation mAP per epoch) and a task on reading them.

Your YOLOv7 workbook

```
# Task: choose training params (see below)
# Task: choose YOLO config params, e.g., augmention, IOU threshold, see file hyp.scratch.cattle.yaml
#### BEGIN SOLUTION
batch_size = ...
epochs = ...
### END SOLUTION
```

Sample Solution

```
#### BEGIN SOLUTION
batch_size = 16
epochs = 10
### END SOLUTION
```

Train

```
# adjust epochs to get a satisfying result
!python train.py \
  --device 0 \
  --workers 8 \
  --batch-size $batch_size \
  --epochs $epochs \
  --img 640 640 \
  --data {data_dir}/cattle.yaml \
  --hyp {data_dir}/hyp.scratch.cattle.yaml \
  --cfg cfg/training/yolov7x.yaml \
  --weights yolov7x.pt \
  --name yolov7x-cattle
```

New

Training Settings: How Long and How Big?

The original YOLO paper trained for hundreds of epochs on COCO's 118,000 images. We have a few hundred images, a pretrained model, and a free Colab GPU with 15 GB of memory.

Before running the code:

- How many epochs do you expect to need, more or fewer than for COCO? Why?

Explore:

1. Set `EPOCHS` and `BATCH_SIZE` in the cell below. `EPOCHS` can be 5 to 50; `BATCH_SIZE` 8, 16 or 32 (a T4 handles 16 at 640 px for YOLO11n). Each extra 10 epochs costs roughly three minutes on a T4.
2. Run the training cell after it.

Observe and interpret:

- Look at the curves after training. Did the model stop improving before the last epoch, or would more epochs help?

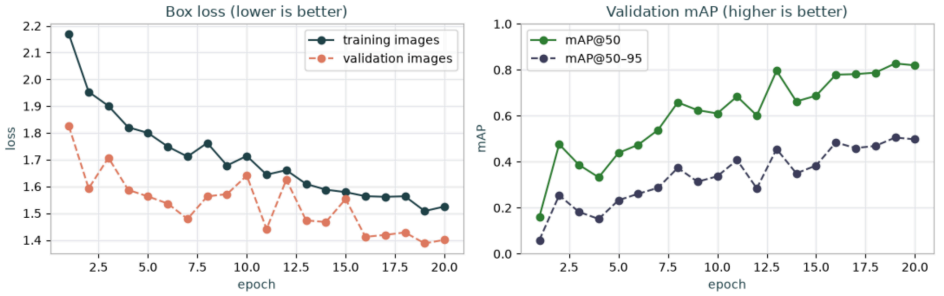
```
# ✎ YOUR TURN: training settings
EPOCHS = 20  # @param {type:"slider", min:5, max:50, step:5}
BATCH_SIZE = 16  # @param [8, 16, 32] {type:"raw"}
IMG_SIZE = 640  # pixels, the size YOLO11 was pretrained at

if BATCH_SIZE not in (8, 16, 32) or not 1 <= EPOCHS <= 100:
    raise ValueError("Choose BATCH_SIZE from 8, 16, 32 and EPOCHS between 1 and 100.")
print(f"Training for {EPOCHS} epochs with batches of {BATCH_SIZE} images at {IMG_SIZE} px.")
```

```
Training for 20 epochs with batches of 16 images at 640 px.
```

New

```
# ▸ 🔎 Plot the Training Curves
```



5 · TP, FP, FN kept; the IoU implementation removed

Your TP/FP/FN task stays, now with every box colored by the matching rule (green found, orange invented, red missed) and Previous / Next through the validation images. The IoU coding task is gone, as you suggested; Know Your Images has the optional IoU task.

Your YOLOv7 workbook

Check the labels and predictions of the sample image.

- What are true positives, false positives, and false negatives?
- Calculate the IOU and print it for the boxes that match with an IOU of at least 0.5

```
def calculate_iou(box1, box2):

    x_min1, y_min1 = box1[0]
    x_max1, y_max1 = box1[1]
    x_min2, y_min2 = box2[0]
    x_max2, y_max2 = box2[1]

    #TODO: continue here
    iou = 0

    return iou
```

New

🟢 True Positives, False Positives, False Negatives: What Went Wrong?

Each image shows the ground truth as dashed outlines and the predictions as solid boxes with their class and confidence. The colors already apply the rule from the box above.

Before running the code:

- For the weed detector, which kind of mistake do you expect more often: invented weeds (FP) or missed weeds (FN)?

Explore:

- Browse at least ten validation images with Previous / Next.

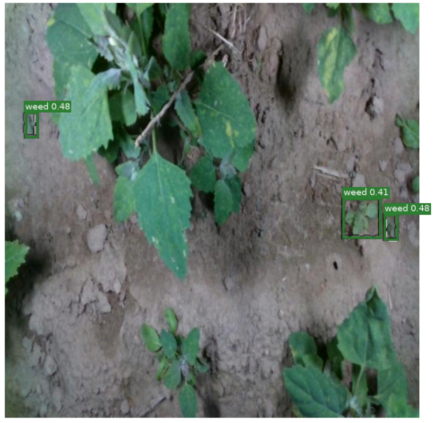
Observe and interpret:

- Find one example of each: a true positive, a false positive, a false negative.
- For the false positives and false negatives: why might the model have made each mistake? Look at object size, occlusion, the image edge, and whether the ground truth itself is right.

```
# ▸ 🔎 Browse Predictions vs. Ground Truth
```

None

12123_jpg.rf.c1da4f4ccf1c2549083c66214bb7d9b7.jpg · TP 3 · FP 0 · FN 0



true positive (found) · false positive (invented) · false negative (missed) · ground truth, matched (white)

6 · New: confidence threshold and precision–recall

A slider moves the confidence threshold; one image shows the TP/FP/FN at that threshold and the precision–recall curve shows where the threshold puts you. Shown here at 0.70.

New in this version

The Confidence Threshold: Precision or Recall?

Imagine the drone sprays every spot where the detector shows a weed box. An invented box wastes herbicide on bare soil, and a missed weed stays in the field and keeps growing.

Before running the code:

- If you raise the confidence threshold from 0.1 to 0.8, what happens to precision and to recall? Write down your guess.

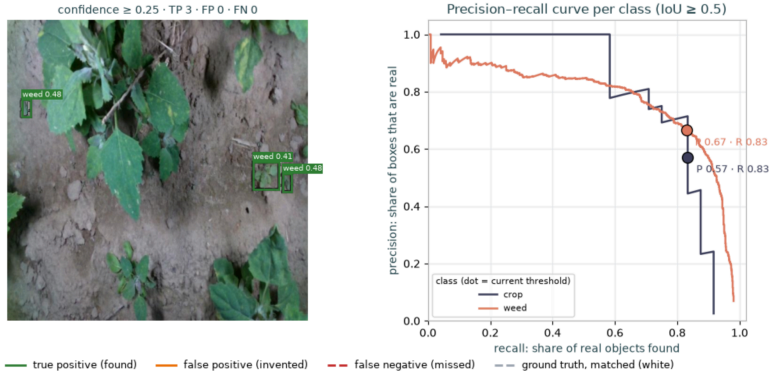
Explore:

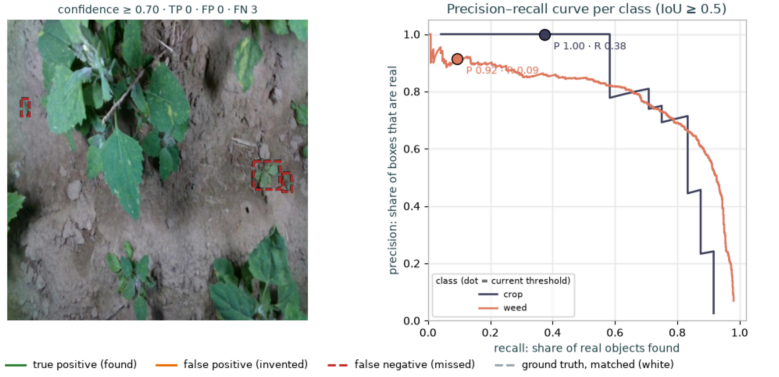
1. Move the `confidence` slider from low to high. The left panel shows one image at that threshold; the right panel shows the precision–recall curve of each class, with a dot where the threshold puts you.
2. Pick two or three other images with the `image` slider and repeat.

Observe and interpret:

- What happens to the orange (invented) and the red (missed) boxes as the threshold rises?
- Which threshold would you choose for spraying, and why is there no threshold that is best for both precision and recall?

```
# ▸ 🔎 Explore the Confidence Threshold
```





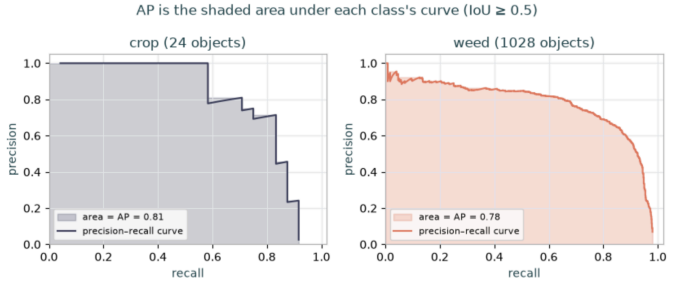
7 · New: AP, mAP@50 and mAP@50–95

AP drawn as the shaded area under each class's curve, checked against Ultralytics' own value; then AP at the ten IoU thresholds whose mean is mAP@50–95, with a task on what each metric rewards.

New in this version

```
# ▸ 🔎 Compute AP per Class and Compare with Ultralytics
```

```
Ultralytics 8.4.208 🚀 Python-3.12.12 torch-2.15.0 MPS (Apple M2 Pro)
YOLO11n summary (fused): 101 layers, 2,597,294 parameters, 0 gradients, 6.4 GFLOPs
val: Fast image access ✅ (ping: 0.0±0.0 ms, read: 1158.4±165.8 MB/s, size: 61.4 KB)
val: Scanning /Users/simone.bonato/Work/notebooks/yolo-object-detection-notebook/data/datasets/weed-crop-aerial-500-150/valid/labels.
cache... 150 images, 0 backgrounds, 0 corrupt: 100% ━━━━━━━━━━━━ 150/150 89.9Mit/s 0.0s
                 Class     Images  Instances      Box(P          R      mAP50  mAP50-95): 100% ━━━━━━━━━━━━ 10/10 5.4it/s 1.8s
                   all        150       1052      0.754      0.789      0.828      0.504
Speed: 0.1ms preprocess, 2.8ms inference, 0.0ms loss, 4.8ms postprocess per image
```




```
mAP@50, our computation: 0.791 (mean of the class APs above)
mAP@50, Ultralytics val:  0.828 (small differences come from how boxes are matched)
```

🟡 mAP@50 vs. mAP@50–95: What Does Each Reward?

So far a prediction counted as correct if its IoU with the ground truth was at least 0.5. A box that covers only two-thirds of an object can pass that bar.
mAP@50–95 repeats the AP computation at ten stricter IoU thresholds, 0.50, 0.55, ..., 0.95, and averages them. It is the main COCO metric and the one in the model table at the top.

Before running the code:

- Will AP go up or down as the required IoU rises from 0.50 to 0.95?

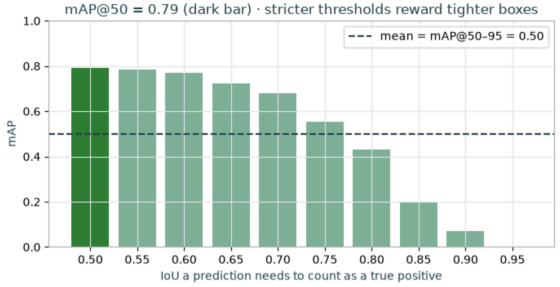
Explore:

- Run the cell. Each bar is the mAP at one IoU threshold; the dashed line is their mean, the mAP@50–95.

Observe and interpret:

- How large is the gap between mAP@50 and mAP@50–95 for our model? What does the gap tell you about the boxes it draws?
- Name one application where mAP@50 is the number to look at, and one where mAP@50–95 matters more.

```
# ▸ 🔎 AP at Ten IoU Thresholds
```

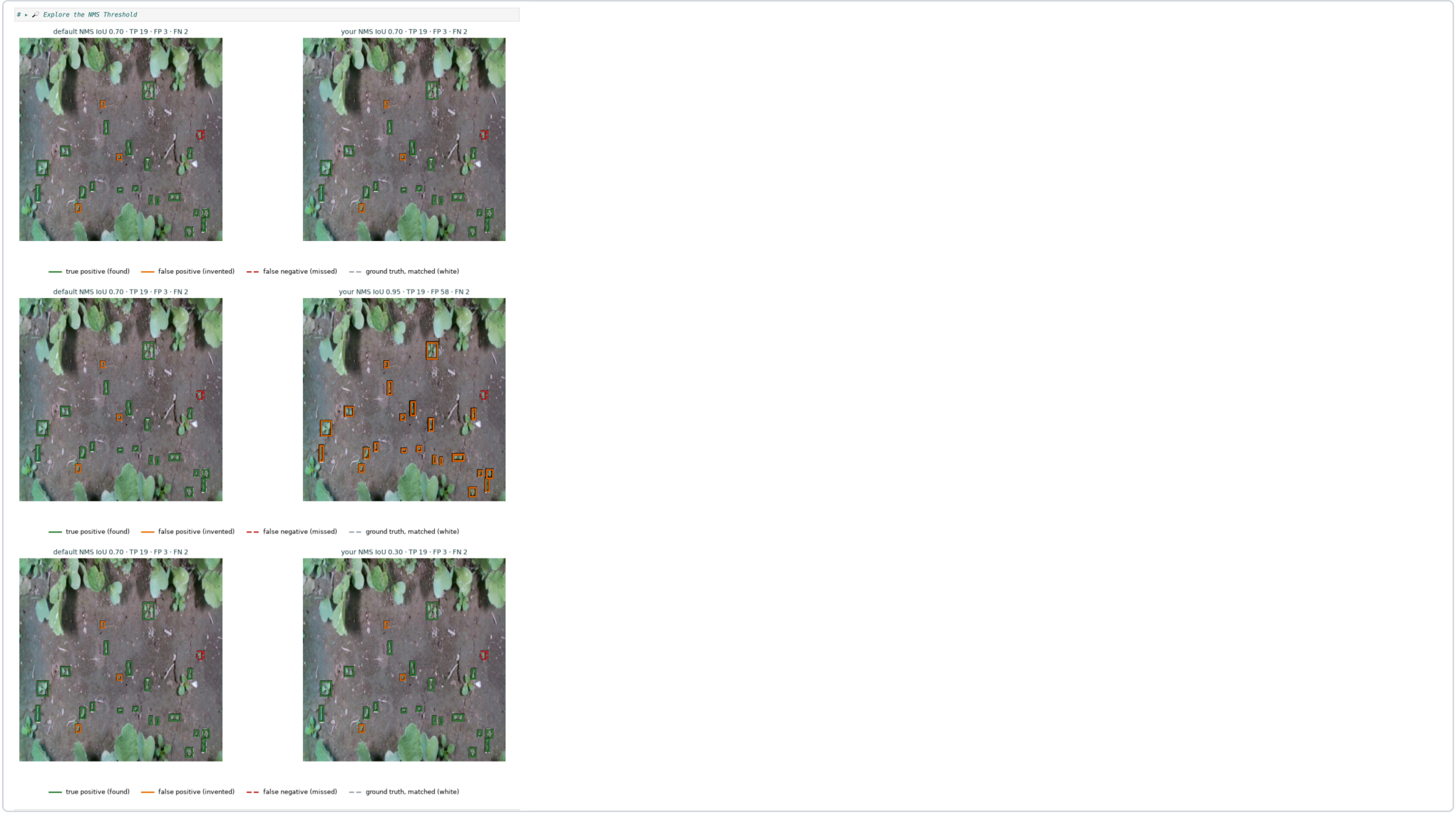



```
mAP@50–95, our computation: 0.501 · Ultralytics val: 0.504
```

8 · New: the NMS slider

The validation image with the most objects, at the default NMS IoU of 0.7 next to the student's setting. Shown at 0.95: duplicates appear (false positives 3 → 58). At 0.3 nothing changes on this dataset, because the weeds rarely overlap; the task and its Sample Answer say so, so the 'lost neighbors' side of NMS is only described here, not seen. A crowded dataset (the aerial-cows variant) would show both sides.

New in this version



9 · YOLOv7-tiny became YOLO11n vs YOLO11s (optional)

Both optional experiments stay behind one checkbox, so Run all stays fast. In our run YOLO11s was about as good as YOLO11n on 500 images (mAP@50 0.82 vs 0.83, mAP@50–95 0.52 vs 0.50) and took 1.6 times as long; freezing the backbone lowered mAP@50 to 0.75.

Your YOLOv7 workbook

1 Optional Exercise: Download and Train Tiny YOLO Model

```
# download trained weights, only needed once or upon updates
!wget https://github.com/WongKinYiu/yolov7/releases/download/v0.1/yolov7-tiny.pt
```

Exercise

Check individual predictions. The results can be found in the `runs/detect/...` folder of `YOLO`.

```
!python train.py --device 0 --workers 8 --batch-size 16 --epochs 200 \
 --img 640 640 --data data/cattle/cattle.yaml \
 --hyp data/cattle/hyp.scratch.cattle.yaml \
 --cfg cfg/training/yolov7-tiny.yaml \
 --weights yolov7-tiny.pt --name yolov7-tiny-cattle
```

New

1 YOLO11n vs. YOLO11s: Is the Larger Model Worth It? (Optional)

YOLO11s has about 3.6 times as many parameters as YOLO11n.

Before running the code:

- How much better do you expect YOLO11s to be on our small dataset, and how much slower?

Explore:

- Run the cell. It trains YOLO11s with your settings and compares both models on the validation images.

Observe and interpret:

- Compare mAP@50, mAP@50–95, training time and prediction time per image.
- For which use would you choose each model?

```
# ▸ 🔎 Train YOLO11s and Compare
```

```
Ultralytics 8.4.208 🚀 Python-3.12.12 torch-2.15.0 MPS (Apple M2 Pro)
engine/trainer: agnostic_nms=False, amp=True, angle=1.0, augment=False, auto_augment=randaugment, batch=16, bgr=0.0, box=7.5, cache=F
alse, cfg=None, channels_last=None, classes=None, close_mosaic=10, cls=0.5, cls_pw=0.0, cls_remap=True, compile=False, conf=None, cop
y_paste=0.0, copy_paste_mode=flip, cos_lr=False, cutmix=0.0, data=/Users/simone.bonato/Work/notebooks/yolo-object-detection-notebook/
data/datasets/weed-crop-aerial-500-150/data.yaml, degrees=0.0, deterministic=True, device=mps, dfl=1.5, dgrad=0.5, dis=6.0, distill_m
odel=None, dlam=1.0, dlog=1.0, dnn=False, dropout=0.0, dynamic=False, embed=None, epochs=20, erasing=0.4, exist_ok=True, fliplr=0.5,
flipud=0.0, format=torchscript, fraction=1.0, freeze=None, hsv_h=0.015, hsv_s=0.7, hsv_v=0.4, imgsz=640, iou=None, kobj=1.0, line_wid
th=None, lr0=0.01, lrf=0.01, mask_ratio=4, max_det=300, mixup=0.0, mode=train, model=yolo11s.pt, momentum=0.937, mosaic=1.0, multi_sc
ale=0.0, name=yolo11s, nbs=64, nms=None, opset=None, optimize=False, optimizer=auto, overlap_mask=True, patience=100, perspective=0.
0, plots=True, pose=12.0, pretrained=True, profile=False, project=/Users/simone.bonato/Work/notebooks/yolo-object-detection-notebook/
runs, quantize=None, rect=False, resume=False, retina_masks=False, rle=1.0, save=True, save_conf=False, save_crop=False, save_dir=/Us
ers/simone.bonato/Work/notebooks/yolo-object-detection-notebook/runs/yolo11s, save_frames=False, save_json=False, save_period=-1, sav
e_txt=False, scale=0.5, seed=42, shear=0.0, show=False, show_boxes=True, show_conf=True, show_labels=True, simplify=True, single_cls=
False, source=None, split=val, stream_buffer=False, task=detect, tile=False, time=None, tracker=tracktrack.yaml, translate=0.1, val=T
rue, verbose=True, vid_stride=1, visualize=False, warmup_bias_lr=0.1, warmup_epochs=3.0, warmup_momentum=0.8, weight_decay=0.0005, wo
rkers=2, workspace=None
Overriding model.yaml nc=80 with nc=2

                   from  n    params  module                                       arguments
  0                  -1  1       928  ultralytics.nn.modules.conv.Conv             [3, 32, 3, 2]
  1                  -1  1     18560  ultralytics.nn.modules.conv.Conv             [32, 64, 3, 2]
  2                  -1  1     26080  ultralytics.nn.modules.block.C3k2            [64, 128, 1, False, 0.25]
  3                  -1  1    147712  ultralytics.nn.modules.conv.Conv             [128, 128, 3, 2]
  4                  -1  1    103360  ultralytics.nn.modules.block.C3k2            [128, 256, 1, False, 0.25]
  5                  -1  1    590336  ultralytics.nn.modules.conv.Conv             [256, 256, 3, 2]
  6                  -1  1    346112  ultralytics.nn.modules.block.C3k2            [256, 256, 1, True]
  7                  -1  1   1180672  ultralytics.nn.modules.conv.Conv             [256, 512, 3, 2]
  8                  -1  1   1380352  ultralytics.nn.modules.block.C3k2            [512, 512, 1, True]
  9                  -1  1    656896  ultralytics.nn.modules.block.SPPF            [512, 512, 5]
 10                  -1  1    990976  ultralytics.nn.modules.block.C2PSA           [512, 512, 1]
 11                  -1  1         0  torch.nn.modules.upsampling.Upsample         [None, 2, 'nearest']
 12             [-1, 6]  1         0  ultralytics.nn.modules.conv.Concat           [1]
 13                  -1  1    443776  ultralytics.nn.modules.block.C3k2            [768, 256, 1, False]
 14                  -1  1         0  torch.nn.modules.upsampling.Upsample         [None, 2, 'nearest']
 15             [-1, 4]  1         0  ultralytics.nn.modules.conv.Concat           [1]
 16                  -1  1    127680  ultralytics.nn.modules.block.C3k2            [512, 128, 1, False]
 17                  -1  1    147712  ultralytics.nn.modules.conv.Conv             [128, 128, 3, 2]
 18            [-1, 13]  1         0  ultralytics.nn.modules.conv.Concat           [1]
 19                  -1  1    345472  ultralytics.nn.modules.block.C3k2            [384, 256, 1, False]
 20                  -1  1    590336  ultralytics.nn.modules.conv.Conv             [256, 256, 3, 2]
 21            [-1, 10]  1         0  ultralytics.nn.modules.conv.Concat           [1]
 22                  -1  1   1511424  ultralytics.nn.modules.block.C3k2            [768, 512, 1, True]
 23        [16, 19, 22]  1    820182  ultralytics.nn.modules.head.Detect           [2, 16, None, [128, 256, 512]]
YOLO11s summary: 181 layers, 9,428,566 parameters, 9,428,550 gradients, 21.7 GFLOPs

Transferred 493/499 items from pretrained weights
Freezing layer 'model.23.dfl.conv.weight'
train: Fast image access ✅ (ping: 0.0±0.0 ms, read: 82.7±16.7 MB/s, size: 60.9 KB)
train: Scanning /Users/simone.bonato/Work/notebooks/yolo-object-detection-notebook/data/datasets/weed-crop-aerial-500-150/train/label
s.cache... 500 images, 0 backgrounds, 0 corrupt: 100% ━━━━━━━━━━━━ 500/500 161.3Mit/s 0.0s
val: Fast image access ✅ (ping: 0.0±0.0 ms, read: 1179.6±226.1 MB/s, size: 70.1 KB)
val: Scanning /Users/simone.bonato/Work/notebooks/yolo-object-detection-notebook/data/datasets/weed-crop-aerial-500-150/valid/labels.
cache... 150 images, 0 backgrounds, 0 corrupt: 100% ━━━━━━━━━━━━ 150/150 78.6Mit/s 0.0s
optimizer: 'optimizer=auto' found, ignoring 'lr0=0.01' and determining best 'optimizer' and 'lr0' automatically...
optimizer: AdamW(lr=0.001667, momentum=0.9) with parameter groups 81 weight(decay=0.0), 88 weight(decay=0.0005), 87 bias(decay=0.0)
Plotting labels to /Users/simone.bonato/Work/notebooks/yolo-object-detection-notebook/runs/yolo11s/labels.jpg...
Using 500 train, 150 val images for fraction=1.0 at imgsz=640
Using 0 dataloader workers
Logging results to /Users/simone.bonato/Work/notebooks/yolo-object-detection-notebook/runs/yolo11s
Starting training for 20 epochs...

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
/Users/simone.bonato/Work/notebooks/yolo-object-detection-notebook/.venv/lib/python3.12/site-packages/torch/autograd/graph.py:1113: U
serWarning: scatter_reduce_mps does not have a deterministic implementation, but you set 'torch.use_deterministic_algorithms(True, wa
rn_only=True)'. You can file an issue at https://github.com/pytorch/pytorch/issues to help us prioritize adding deterministic support
for this operation. (Triggered internally at /Users/runner/work/pytorch/pytorch/aten/src/ATen/Context.cpp:224.)
```